

Hardware

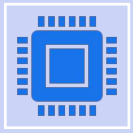
YEAR 9 COMPUTING TECHNOLOGY

Learning Intention

- Investigate how hardware and software are used to manage, control and secure the movement of, and access to, data in networked digital systems



Success Criteria



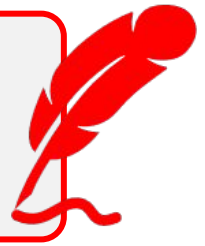
I can differentiate between input and output devices.



I can describe hardware components and storage devices used for computing.

What Is Hardware?

Hardware: physical components of a computer system



Computers are **information processors** – data is input, processed, stored, or displayed

Hardware components work together to complete tasks

Main functions: **input, output, storage, processing, control**

Input Devices



Collect data from the outside world and convert it for the computer



Common devices: keyboard, mouse, touchpad, microphones, scanners, cameras, graphics tablets



Sensors can collect data like temperature, light, sound, and movement



Input data is stored in memory until processed by the CPU

Output Devices



Communicate information from computer to user



Common devices: monitor, printer, speakers



Specialised output: 3D printers, laser cutters, head-up displays

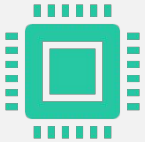


Monitors convert digital signals to visual patterns



Speakers convert digital signals to sound waves

Storage



Primary storage: short-term, volatile (RAM, cache)



Secondary storage: long-term, non-volatile

Magnetic: hard drives

Optical: CDs, DVDs

Solid-state/flash: USB drives, SSDs



Storage can be **local** or **remote (cloud storage)** for accessibility and security

Processing and Control



CPU: often called the “brain” the primary chip on the motherboard responsible for interpreting and executing instructions from software and hardware

Computer chips are made of **microprocessors** (millions of transistors)

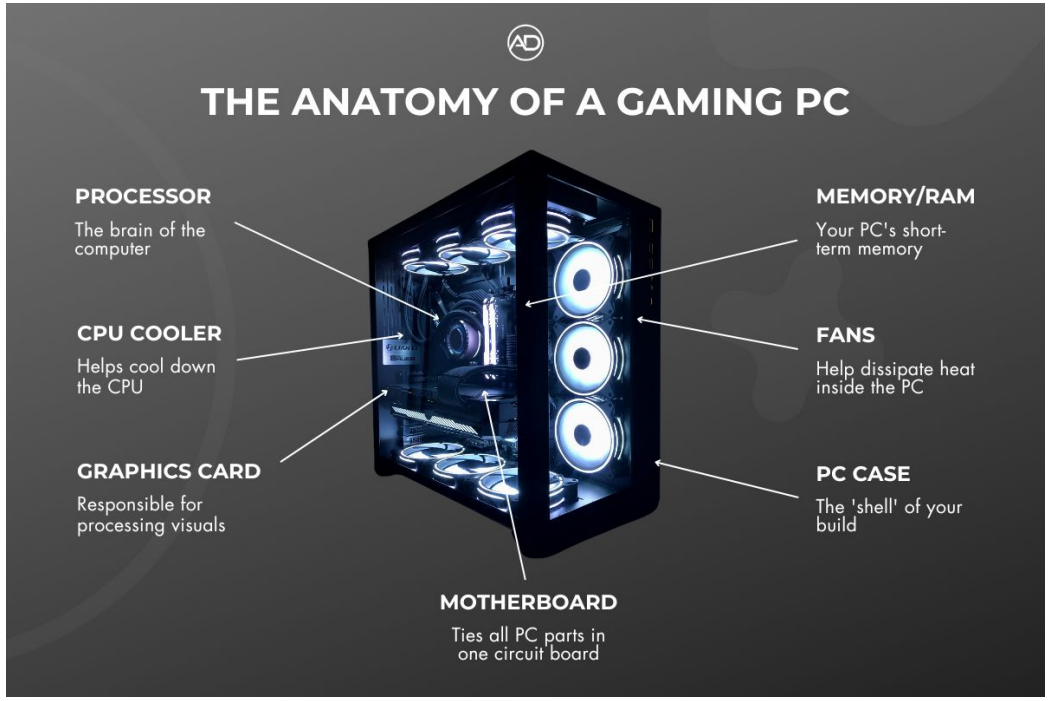
RAM: volatile, temporary storage

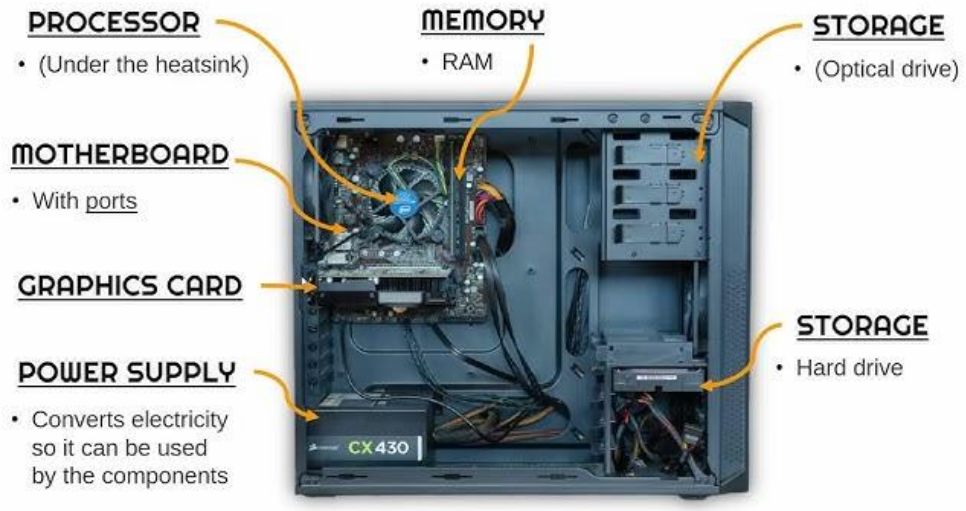
ROM: permanent storage, contains essential startup information

GPU: handles graphics tasks, video rendering, AI, and parallel processing tasks

Chips work together to process, store, and control data

Inside a PC


THE ANATOMY OF A GAMING PC
PROCESSOR
The brain of the computer
CPU COOLER
Helps cool down the CPU
GRAPHICS CARD
Responsible for processing visuals
MEMORY/RAM
Your PC's short-term memory
FANS
Help dissipate heat inside the PC
PC CASE
The 'shell' of your build
MOTHERBOARD
Ties all PC parts in one circuit board


PROCESSOR
• (Under the heatsink)
MEMORY
• RAM
STORAGE
• (Optical drive)
MOTHERBOARD
• With ports
GRAPHICS CARD
POWER SUPPLY
• Converts electricity so it can be used by the components
STORAGE
• Hard drive
CX430

Classification of Computer Hardware



Smartphones: portable, many sensors, mostly server-processed



Personal computers: self-contained, handle input, processing, output, storage

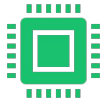
Desktops, laptops, workstations



Servers: networked, manage software, files, and access for multiple users



Mainframes: large-scale, handle massive transactions (banks, government)



Supercomputers: fastest, perform trillions of calculations (weather, science)



Quantum computers: use qubits, perform complex calculations at unprecedented speeds

RAM and Secondary Storage

